Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 18)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Give **two** wild species with potential as new food species and, for each, one characteristic that makes it useful.

[2 marks]

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0 1 . 2 Explain why indigenous species may be better for domestication than introduced livestock or crops.

[2 marks]

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0 1 . 3 Explain how species of the spinach family could help to solve a problem caused by irrigation.

[1 mark]

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0 2 . 1 Explain why carnivorous species such as salmon dominate aquaculture in MEDCs while herbivores such as carp and tilapia dominate in LEDCs.

[3 marks]

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0 2 . 2 Describe how all-female rainbow trout are produced and explain why this is done.

[3 marks]

0 2 . 3 Explain how each of the following abiotic factors is controlled in an intensive salmon farm.

[4 marks]

Dissolved oxygen

Water flow

Day length

Temperature

10

0 3 . 1 Explain the effects of deforestation on soil.

[4 marks]

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0 3 . 2 Explain how deforestation affects hydrology and climate.

[4 marks]

0 3 . 3 Give **two** ecological impacts of deforestation other than the direct loss of species.

[2 marks]

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0 4 . 1 Describe the main principles of the circular economy.

[6 marks]

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0 4 . 2 Explain what is meant by 'optimum production rather than maximum production', using a comparison between a natural ecosystem and an agroecosystem.

[2 marks]

0 4 . 3 Explain why international cooperation (eg through the EU or UN) is needed to implement circular economy initiatives.

[2 marks]

10

0 5 Table 1 describes three genetically modified (GM) crops.

Table 1

GM crop	Gene source	Characteristic
Bt maize / cotton	Bacillus thuringiensis bacterium	Produces a toxin that kills insect pests
Roundup Ready soya	Agrobacterium	Resistant to the herbicide glyphosate
Golden rice	Daffodil and Erwinia bacterium	Produces vitamin A in the grain

0 5 . 1 Explain the main advantage of GM over conventional selective breeding.

[2 marks]

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0 5 . 2 For each crop in Table 1, explain one potential benefit.

[3 marks]

Bt crops .....

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Roundup Ready soya .....

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Golden rice .....

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0 5 . 3 Explain **four** concerns about GM crops.

[4 marks]

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0 5 . 4 Give **one** reason why GM crops have been slow to enter cultivation in Europe.

[1 mark]

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0 6 . 1 Explain how natural material cycles differ from linear human systems of material use.

[4 marks]

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0 6 . 2 Explain why the use of toxic metals in electronic products is a particular problem for sustainability.

[2 marks]

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0 6 . 3 Describe how the design of a product can move it from a linear to a cyclical system.

[4 marks]

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0 7 Table 1 compares two exploited species.

Table 1

Feature	Anchovy	Blue whale
Age at sexual maturity	1 year	10 years
Offspring per female per year	Thousands of eggs	1 calf every 2-3 years
Lifespan	3-4 years	80+ years
Parental care	None	Extensive

0 7 . 1 Use Table 1 to identify which species is r-selected and which is k-selected, and explain the difference between the strategies.

[3 marks]

0 7 . 2 Explain why k-selected species are much more easily over-exploited than r-selected species.

[3 marks]

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0 7 . 3 State the equation used to forecast a change in population size.

[1 mark]

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0 7 . 4 Explain what is meant by **maximum sustainable yield** and how it relates to carrying capacity.

[3 marks]

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0 8 Table 1 shows the number of plant species per m^2 in a chalk grassland reserve under different management, ten years after management began.

Table 1

Management	**Mean plant species per m^2**	**Percentage of area covered by scrub / bushes**
No management (abandoned)	8	55
Sheep grazing	24	5
Annual mowing (August)	19	0
Winter burning	14	10

0 8 . 1 Explain what is meant by a **plagioclimax** community and why chalk grassland is an example.

[2 marks]

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0 8 . 2 Use Table 1 to explain why grazing produced the highest plant diversity.

[4 marks]

0 8 . 3 Explain why different grazing animals are chosen for different plagioclimax habitats.

[2 marks]

0 8 . 4 Explain how burning maintains heathland.

[2 marks]

10

0 9 . 1 Which of the following is an example of habitat destruction rather than a change in abiotic factors? Tick one box.

[1 mark]

A Thermal pollution of a river by a power station ☐

B Flooding a valley to create a reservoir ☐

C Lowering the water table by groundwater abstraction ☐

D Acid mine drainage entering a stream ☐

0 9 . 2 Give **four** human activities that cause complete habitat destruction.

[4 marks]

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1 0 . 1 Explain how the relationship between coral polyps and zooxanthellae allows a high biomass of life in nutrient-poor tropical seas.

[4 marks]

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1 0 . 2 Explain why growing coral reefs help to resist climate change.

[2 marks]

1 0 . 3 Discuss the importance of tropical coral reefs and evaluate the methods used to conserve them.

[9 marks]

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Extra space

15

Write an essay on one of the following topics.

1 1 . 1 Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness.

[25 marks]

OR

1 1 . 2 Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS